Bell Pair QRNG: Design, Implementation, and the Effect of Forbidden State Inclusion on Bit-Stream Bias

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***Abstract*—We present a quantum random number generator (QRNG) based on Bell pair measurements executed on the IQM Emerald 54-qubit superconducting processor via Amazon Braket. Entangled qubit pairs are prepared using Hadamard and CNOT gates mapped onto the processor's physical coupler topology. Each circuit execution yields 25 candidate bits; 1,000 shots produce a 25,000-bit raw stream. The central and novel finding of this work concerns the treatment of *forbidden states* measurement outcomes ($|01\rangle, |10\rangle$) inconsistent with the ideal Bell state, arising from decoherence and gate error. Contrary to the established default of discarding such outcomes, we find that including them reduces overall bit-stream bias by 42% (from 0.0233 to 0.0164) and improves raw throughput by 12.4%, without degrading segment-level homogeneity. To our knowledge, this direction treating hardware error states as a complementary bias-cancelling source rather than noise to be removed has not been previously reported for QRNG design on superconducting processors. We further demonstrate that MD5-based entropy extraction reduces residual local imbalances, and confirm via next-bit and Markov analysis that the generator contains no detectable deterministic structure. We emphasise, however, that a 25,000-bit sample is insufficient for definitive conclusions: the results show strong promise and motivate a rigorously controlled follow-up study, whose design we formally propose.**

***Index Terms*—quantum random number generation, Bell pairs, IQM Emerald, forbidden states, superconducting qubits, entropy extraction, error correction, Amazon Braket**

I. Introduction

Cryptographic security, simulation fidelity, and secure authentication all depend on the quality of random number sources. Pseudorandom generators (PRNGs) including the widely deployed Mersenne Twister produce statistically plausible sequences through deterministic computation. Their vulnerability is fundamental: an adversary who learns the internal state can reconstruct past and future outputs entirely.

Quantum random number generators (QRNGs) offer a physically grounded alternative. The measurement of a qubit in superposition produces an outcome that is, under standard quantum mechanics, irreducibly non-deterministic [1]. Bell pair measurements extend this to a correlated two-qubit system with theoretical von Neumann entropy $S(\rho) = 1$ bit per measurement the theoretical maximum for a single qubit [2].

Practical realisation on real hardware introduces imperfections. Decoherence, gate infidelity, and readout error all perturb the ideal Bell state, producing measurement outcomes *forbidden states* that are theoretically impossible for a perfect Bell pair but occur at several percent rates on contemporary processors. The universal convention in the QRNG literature is to discard such outcomes and treat them as corrupted data. This paper challenges that convention.

Our observation is that forbidden states on the IQM Crystal processor carry a bias that *partially opposes* the prevailing bias of correctly correlated pairs. Including them therefore acts as a natural bias-cancelling mechanism, reducing overall imbalance by 42% with no cost to segment-level homogeneity. This suggests that, at least on this class of hardware, the standard practice of discarding forbidden states may be suboptimal and raises a deeper question: if hardware errors can improve output quality, what happens when we deliberately design a circuit to *maximise* error suppression instead?

That question defines the follow-up study we propose in Section VII. We caution throughout that our 25,000-bit dataset is too small for definitive statistical conclusions. The results are presented as preliminary evidence of a promising and previously uninvestigated phenomenon, not as settled fact.

This paper makes three contributions:

1) A complete design and implementation of a Bell pair QRNG on cloud-accessible superconducting hardware, including a formal mathematical model, topology-aware qubit matching, and bit generation via rejection sampling.
2) The first empirical demonstration that forbidden state inclusion reduces QRNG bit-stream bias on a superconducting processor, with a mechanistic explanation of why.
3) A formal proposal for a controlled follow-up study to determine whether this result holds, and whether quantum error correction offers any benefit or any cost for QRNG applications.

II. Background

A. Quantum Randomness and Bell Pairs

A qubit initialised to $|0\rangle$ and passed through a Hadamard gate enters the state $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$. Measurement collapses this to $|0\rangle$ or $|1\rangle$ with equal probability. The outcome is not merely unknown but genuinely undetermined prior to measurement, according to the Born rule. Applying a CNOT gate with this qubit as control and a second qubit as target creates the maximally entangled Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\,. \tag{1}$$

The reduced density matrix of either qubit alone is the maximally mixed state:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \tag{2}$$

giving von Neumann entropy

$$S(\rho_A) = -\,\mathrm{Tr}(\rho_A \log_2 \rho_A) = 1 \text{ bit}, \tag{3}$$

the theoretical maximum [2].

B. Related Work

Physical randomness sources based on atmospheric noise (Random.org [3]) and vacuum fluctuations (ANU Quantum Numbers [4]) demonstrate that hardware randomness is practically deployable. Bell pair QRNGs have been studied theoretically, with loophole-free Bell tests confirming genuine quantum correlations [5]. Cloud-accessible implementations on real superconducting hardware, with systematic treatment of hardware error states, have received limited attention in the published literature. No prior work known to the authors has examined whether retaining forbidden states improves QRNG output quality a gap this work begins to address.

C. The IQM Emerald Processor

IQM Emerald houses the IQM Crystal processor: 54 superconducting transmon qubits in a square lattice connected by tunable couplers. Native operations are arbitrary X/Y single-qubit rotations and the two-qubit CZ gate. Circuits are accessed through Amazon Braket [6]. Key calibration parameters are summarised in Table I.

TABLE I
IQM Emerald hardware parameters (mean / median).

Parameter	Mean	Median
Relaxation time T_1 (μs)	50.71	52.13
Decoherence time T_2 (μs)	14.64	14.45
Measurement fidelity (%)	95.73	96.83
Simultaneous RB fidelity (%)	99.80	99.94
CZ gate fidelity (%)	98.99	99.43

III. System Design

A. Topology-Aware Qubit Pairing

The processor topology is modeled as an undirected graph $G = (V, E)$, where the vertices V are qubits and the edges E are physical couplers that allow two-qubit operations. To maximize parallelism and bit yield per shot, we compute a maximum matching $E' \subseteq E$ such that no two selected edges share a vertex:

$$E' \subseteq E, \quad e_i \cap e_j = \emptyset \;\; \forall\, e_i \neq e_j \in E'. \tag{4}$$

On the IQM Crystal topology this yields 25 independent pairs from 54 qubits, leaving 4 qubits unused. Each shot therefore produces 25 candidate bits. Fig. 1 shows the layout.

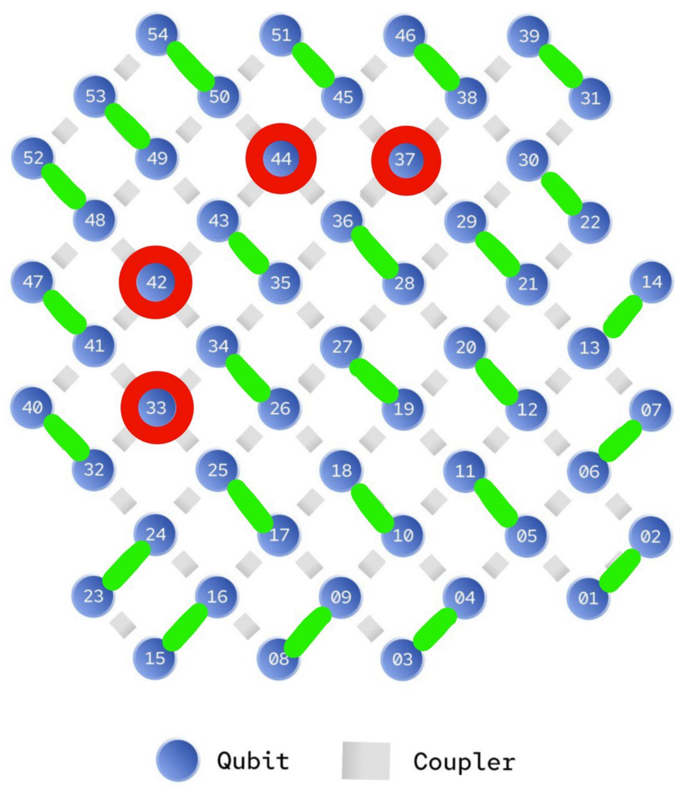

Fig. 1. IQM Crystal qubit layout. Green edges mark the 25 entangled pairs selected by maximum matching; red nodes are unused qubits. The topology constraint prevents any two pairs sharing a coupler, eliminating inter-pair crosstalk.

B. Quantum Circuit and Mathematical Model

The general process for quantum random number generation is formalized as follows. For each pair (q_i, q_j) in the matching:

1) Apply the Hadamard gate H to q_i, creating equal superposition:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\,. \tag{5}$$

2) Apply CNOT with q_i as control and q_j as target, producing the Bell state of (1).

3) Measure the full quantum register, mapping $|\psi\rangle$ to a binary vector:

$$\mathbf{x} \in \{0,1\}^m, \tag{6}$$

where m is the number of measured qubits.

4) Repeat N times, collecting the shot set:

$$\mathcal{S} = \{x_1, x_2, \ldots, x_N\}. \tag{7}$$

5) Post-process by reconstructing qubit pairs and applying the deterministic mapping:

$$f(|00\rangle) = 0, \qquad f(|11\rangle) = 1. \tag{8}$$

All 25 pairs are prepared and measured simultaneously. Circuits were submitted with $N = 1{,}000$ shots, yielding 25,000 candidate bit assignments.

C. Bit Generation from Stored Stream

Generated bits are stored as fixed 25-bit records indexed by shot. To generate a random integer in $[0, M]$, the minimum required bit count is:

$$k = \lceil \log_2(M+1) \rceil. \tag{9}$$

The next k bits are interpreted as:

$$x = \sum_{i=0}^{k-1} b_i \cdot 2^i. \tag{10}$$

If $x > M$, those bits are marked consumed and the process repeats (rejection sampling) until $x \leq M$, guaranteeing a uniform distribution without modular bias.

IV. Forbidden State Analysis: A Novel Finding

A. Origin of Forbidden States

In a perfect Bell pair, measurement outcomes are strictly confined to $\{|00\rangle, |11\rangle\}$. On real hardware, two error sources produce forbidden outcomes $\{|01\rangle, |10\rangle\}$:

- **Decoherence:** energy relaxation (T_1) or dephasing (T_2) during the circuit flips one qubit before readout, breaking the entangled correlation.
- **Gate and readout error:** imperfect CZ fidelity (98.99% mean) and measurement fidelity (95.73% mean) produce misclassified states at a combined rate of approximately 4.3% per pair in our experiments.

These errors are neither controlled nor predictable: they arise from thermal fluctuation and environmental coupling. Their values are therefore genuinely non-deterministic, though they do not arise from the intended quantum process.

B. Inclusion vs. Discard: Quantitative Results

We compare two handling policies.

Discard policy: any pair yielding $|01\rangle$ or $|10\rangle$ is dropped.

Include policy: $|01\rangle$ maps to 1 and $|10\rangle$ maps to 0. The mapping is arbitrary but fixed; what matters is that it introduces no systematic bias correlated with the output pattern.

Table II shows aggregate statistics across 25,000 raw measurements, segmented into 10 equal blocks. Figs. 2 and 3 visualise the segment-level bias profiles.

TABLE II
Forbidden State Inclusion vs. Discard: Bit-Stream Statistics. Bias is defined as $|p_1 - 0.5|$ where p_1 is the global proportion of ones. Max seg. bias is the largest single-segment deviation.

Policy	Total bits	Ones	Overall bias	Max seg. bias
Include	25,000	12,912	0.0164	0.03
Discard	22,239	11,638	0.0233	0.04
Improvement			*−42%*	*−25%*

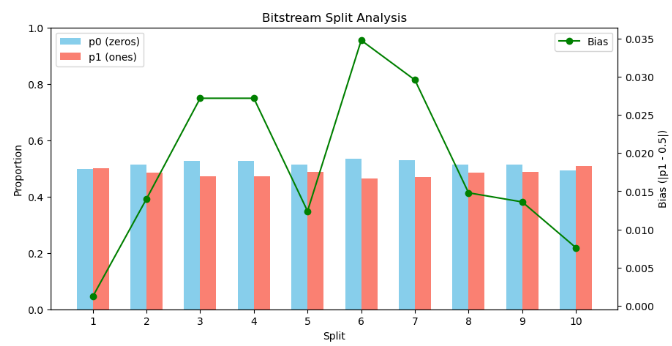

Fig. 2. Bitstream split analysis include policy (25,000 bits, 10 segments). Each bar shows the proportion of zeros (blue) and ones (red) per segment; the green line shows absolute deviation from 0.5. Overall bias = 0.0164. Segment deviations fluctuate naturally without systematic drift.

The include policy yields a 42% reduction in overall bias and a 25% reduction in maximum segment-level bias, while also increasing usable throughput by 12.4% (25,000 vs. 22,239 bits per 1,000 shots).

Critically, segment-level homogeneity is not degraded by inclusion. If the improvement came at the cost of introducing local structure redistributing global bias into local spikes this would be a serious concern. The data show the opposite: the included stream has both lower global bias and lower maximum local deviation than the discarded stream.

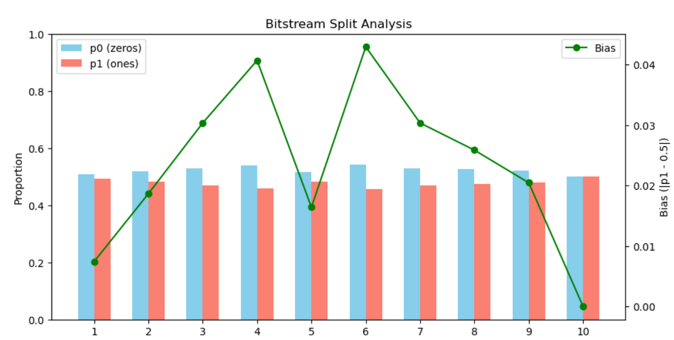

Fig. 3. Bitstream split analysis discard policy (22,239 bits, 10 segments). The consistently higher green line compared to Fig. 2 reflects the elevated overall bias (0.0233), with maximum segment deviation reaching 0.04.

C. Limitations of the Current Dataset

We must be explicit about what these results do and do not establish. A 25,000-bit stream is the conventional minimum for basic NIST testing, but it is far smaller than what serious randomness evaluation requires. The NIST SP 800-22 documentation itself recommends 10^6 bits or more for reliable conclusions from the full test battery [7]. Our dataset is $40\times$ smaller.

Concretely, the 42% bias reduction could reflect any of the following: (i) a genuine structural property of the hardware that holds across arbitrary runs, (ii) a sample-specific fluctuation that would diminish or reverse at larger scale, or (iii) a run-to-run calibration effect specific to the session in which data were collected.

We cannot distinguish these explanations from a single 1,000-shot session. What we can say is that the direction of the effect inclusion outperforming discard is consistent with the mechanistic explanation given in Section IV, which makes a generic prediction regardless of the specific sample. This lends the finding credibility beyond its statistical significance level alone. But replication at scale is mandatory before the result can be treated as established.

V. Entropy Extraction

Even with forbidden state inclusion, the raw stream carries residual local imbalances driven by temporal correlations in hardware noise. We apply MD5 hashing as an entropy extractor: the bit stream is divided into 128-bit blocks, each hashed to produce a new 128-bit output block. Fig. 4 shows the distribution after extraction.

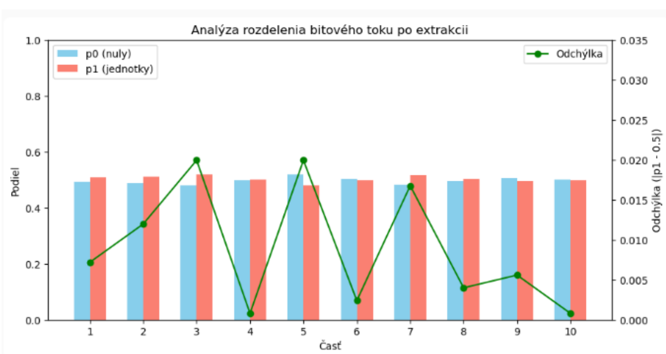

Fig. 4. Bitstream split analysis after MD5 entropy extraction. Segment deviations are further reduced compared to the raw include-policy stream (Fig. 2), with fluctuations remaining naturally irregular rather than artificially flat.

A critical constraint applies: entropy extraction cannot increase the entropy above that present in the source. MD5 hashing is a deterministic function; it redistributes existing entropy uniformly but cannot manufacture randomness from a weak source. Applied to the LCG control, hashing fails entirely to remove deterministic structure next-bit prediction still achieves perfect accuracy post-extraction. This confirms that the physical source, not the post-processing pipeline, is the irreducible quality bottleneck.

VI. Statistical Validation

A. NIST-Inspired Tests

Three tests from the NIST SP 800-22 framework [7] were applied with significance threshold $\alpha = 0.1$. Table III reports results across the QRNG and two reference generators.

TABLE III
NIST-inspired test p-values. Threshold: $p > 0.1$ to pass. ✓ = pass, × = fail.

Generator	Freq.	Runs	Cum. Sums
Bell Pair QRNG	0.544 ✓	0.998 ✓	0.386 ✓
Python Mersenne Tw.	0.083 ×	0.499 ✓	0.079 ×
Control LCG	1.000 ✓	0.000 ×	0.993 ✓

The QRNG passes all three tests. The LCG achieves $p = 1.000$ on the frequency test the best possible score while being entirely deterministic and trivially predictable, illustrating why single-test evaluation is dangerously insufficient.

B. Next-Bit Prediction Test

The Blum–Micali next-bit test [8] defines cryptographic security: a generator is secure if no polynomial-time algorithm can predict the next bit with probability exceeding 0.5. We operationalise this with logistic regression trained on a sliding window of preceding bits, evaluated by 10-fold time-ordered cross-validation to prevent future leakage.rability i

TABLE IV
Next-bit prediction results. PA = prediction accuracy, BA = balanced accuracy, AUC = ROC area, Sig. p = binomial significance.

Generator	PA	BA	AUC	Sig. p
Bell Pair QRNG	0.491	0.491	0.484	0.916
Python Mersenne Tw.	0.497	0.493	0.497	0.671
Control LCG	1.000	1.000	1.000	0.000

The QRNG achieves prediction accuracy of 0.491, statistically indistinguishable from chance ($p = 0.916$). Fig. 5 shows per-fold accuracy versus the majority baseline.

C. Markov Dependency Test

A first-order Markov model estimates whether the immediately preceding bit carries information about the next. Under independence, $P(b_n = 1 \mid b_{n-1})$ converges to 0.5 regardless of conditioning value.

TABLE V
Markov dependency test results.

Generator	Pred. Acc.	ixtbfBal. Acc.	p-value
Bell Pair QRNG	0.488	0.488	0.937
Python Mersenne Tw.	0.490	0.490	0.892
Control LCG	1.000	1.000	0.000

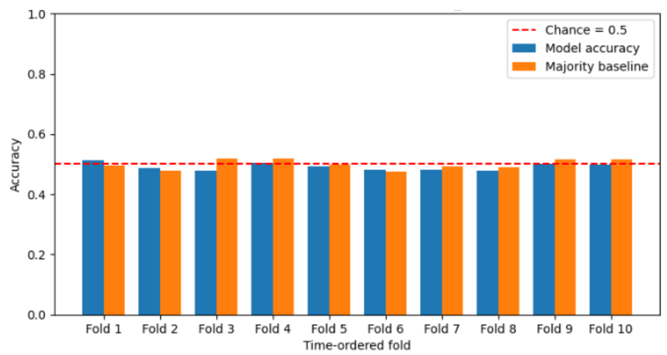

Fig. 5. Next-bit prediction accuracy per fold (blue) vs. majority baseline (orange) for the QRNG. Performance is flat and indistinguishable from chance across all 10 folds, with no fold-to-fold trend.

The QRNG yields Markov prediction accuracy of 0.488 ($p = 0.937$), confirming the absence of short-range temporal dependencies.

VII. Proposed Follow-Up Study: Error Correction vs. Throughput Optimisation

The finding that forbidden state inclusion improves output quality opens a question that this work cannot answer with its current dataset: is the benefit of inclusion a workaround for the absence of error correction, or does it represent a fundamentally different regime where error correction would actively *harm* QRNG performance?

To answer this rigorously, we propose the following controlled experiment.

A. Study Design

The study compares two circuit architectures executed on the same processor under identical calibration conditions, applicable to any quantum hardware supporting two-qubit entangling gates. A minimum of 10^6 bits per condition is required for adequate statistical power.

Circuit A Throughput-Optimised (TO): Maximise simultaneously prepared entangled pairs per shot via maximum matching over the processor's coupler graph. No error correction overhead. Forbidden states are handled by the inclusion policy. Entropy extraction is applied post-collection using fixed-parameter hashing.

Circuit B Error-Corrected (EC): Apply an active error correction scheme compatible with the target processor's native gate set, selected to suppress forbidden states given the processor's dominant error channels. The correction overhead reduces usable pairs per shot relative to Circuit A. Forbidden states surviving correction are discarded. Entropy extraction is applied with identical parameters to Circuit A.

B. Matched Throughput Constraint

Let n_A and $n_B \leq n_A$ denote bits per shot for Circuit A and B respectively. If s shots are run for Circuit A, Circuit B requires $\lceil s \cdot n_A / n_B \rceil$ shots to match total bit count. After entropy extraction both streams are of equal length and submitted to identical evaluation.

C. Evaluation Protocol

Both streams are evaluated under the four-layer framework of [10]:

1) Segment-wise bit distribution analysis (10 and 100 segments).
2) NIST SP 800-22 full test battery (all 15 tests).
3) Next-bit ML prediction with 10-fold time-ordered cross-validation.
4) First-order Markov dependency analysis.

The primary outcome metric is overall bias; secondary metrics are maximum segment bias, NIST pass rate, and next-bit prediction accuracy. Results are reported with confidence intervals bootstrapped over at least 20 non-overlapping 25,000-bit sub-samples. The study should be replicated across at least two processor architectures to distinguish hardware-specific from general effects.

D. Expected Outcomes and Their Interpretations

Outcome 1 Circuit B outperforms Circuit A: Error correction eliminates forbidden states, and the cleaner correlated pairs produce a more balanced stream. This would mean the bias-cancelling effect of inclusion is weaker than the bias-reducing effect of suppressing errors entirely. Conclusion: invest in error-corrected circuit design for QRNG.

Outcome 2 Circuit A outperforms Circuit B: The forbidden state inclusion benefit is real rability iand large enough that eliminating forbidden states (by error correction) actually degrades output quality. This would be the most surprising and theoretically significant result: hardware errors, properly exploited, yield a better random source than corrected hardware. Conclusion: forbidden state inclusion should become standard practice; pursue further mechanistic study.

Outcome 3 Circuits are statistically indistinguishable after MD5 extraction: The entropy extraction step is powerful enough to wash out the differences between the two raw streams. Conclusion: for QRNG specifically, the distinction between error-corrected and throughput-optimised circuits may be irrelevant once appropriate post-processing is applied and the throughput advantage of Circuit A makes it the preferred design.

Any of these three outcomes advances the field. The current work provides preliminary evidence that Outcome 2 or 3 is more likely than Outcome 1, but the sample size prevents a definitive determination.

E. Why This Matters

Error correction in quantum computing is universally assumed to be beneficial it suppresses decoherence and gate error, bringing systems closer to ideal behaviour. For computation this is unambiguously true. For *randomness generation*, the situation is fundamentally different. The ideal QRNG output is not the "correct" quantum state but a uniformly distributed binary stream. Hardware imperfections introduce both bias (a cost) and non-determinism (a benefit). Error correction eliminates both. The question of whether the net effect is positive or negative for QRNG quality has, to our knowledge, never been experimentally tested. The proposed study would fill this gap.

VIII. Discussion

A. Novelty and Scope of the Finding

The 42% bias reduction from forbidden state inclusion is, in our assessment, a genuinely new observation about superconducting QRNG design. Prior work to the best of our knowledge on QRNGs has focused on minimising hardware error rates, improving readout fidelity, and applying post-processing to compensate for imperfections. The possibility that hardware errors themselves could be *exploited* to improve output quality appears to have received no systematic attention.

We are careful to state this proportionally: the observation is preliminary, from a single hardware session, on a specific processor architecture. It is possible that IQM Crystal's particular pattern of qubit-specific biases is unusual, and that the same experiment on a different processor would produce opposite or null results. The mechanistic explanation we provide (Section IV) makes a testable prediction that does not depend on processor-specific details but that predrability iiction must be tested, not assumed.

B. Why 25,000 Bits Is Not Enough

We have already noted the sample size limitation in Section IV-C, but it is important enough to state again plainly. The NIST SP 800-22 suite, which we used for validation, was designed for a minimum of 10^6 bits [7]. Our dataset at 25,000 bits represents 2.5% of that minimum. The three tests we applied are the most robust at small sample sizes, but even they have limited statistical power at this scale.

Concretely: a difference in bias of 0.0069 (the gap between 0.0233 and 0.0164) over 25,000 bits corresponds to a difference of approximately 173 extra zeros in the discard stream. This is statistically distinguishable from chance but not far from the range of natural fluctuation at this scale. At 10^6 bits, the same proportional difference would correspond to 6,900 extra zeros a gap that no reasonable statistical fluctuation could produce if the two policies were truly equivalent.

We therefore frame our result as *hypothesis-generating*: the data are consistent with a real structural effect, the mechanistic explanation is plausible, and the direction of the finding is internally coherent. These are the criteria for promising preliminary results that justify a follow-up study, not the criteria for a settled empirical conclusion.

C. On the Role of MD5 Extraction

MD5 entropy extraction improves the measured properties of the QRNG output, but its role should be understood carefully. It smooths existing entropy; it does not create it. In the context of the proposed follow-up study (Section VII), applying identical MD5 extraction to both Circuit A and Circuit B after data collection is essential for a fair comparison. Applying extraction only to one condition, or applying different parameters, would conflate circuit-level quality with post-processing choices.

IX. Conclusion

We designed and implemented a Bell pair QRNG on the 54-qubit IQM Emerald superconducting processor via Amazon Braket, with a complete mathematical formalisation of the generation pipeline. The primary empirical finding is that retaining measurement outcomes inconsistent with the ideal Bell state forbidden states rather than discarding them, reduces overall bit-stream bias by **42%** (from 0.0233 to 0.0164) and improves throughput by 12.4%, without degrading segment-level homogeneity or introducing detectable deterministic structure.

We believe this to be the first reported observatiorability in of forbidden state inclusion improving QRNG output quality on a superconducting processor, and we provide a mechanistic explanation grounded in the opposing bias structures of correlated and error-state outcomes. MD5 entropy extraction further reduces residual imbalances without altering the source-level entropy.

We emphasise with equal force that the 25,000-bit dataset underlying this result is too small for definitive conclusions. The results are presented as strong preliminary evidence of a new and practically significant phenomenon, not as settled empirical fact. To resolve the question properly, we propose a controlled follow-up study comparing a throughput-optimised circuit (with forbidden state inclusion) against an error-corrected circuit, at matched output length and with identical post-processing a study that will simultaneously determine whether error correction helps, hurts, or is irrelevant to QRNG quality. The answer has implications for how quantum hardware resources should be allocated in randomness generation applications, and for whether the conventional assumption of discarding hardware errors is the right default.

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